

CSE470 Assignment 2

Question 1

- A. Write one single GPT prompt that asks the LLM to generate a scenario where all three design patterns (Observer, Adapter and Singleton) will be used. Then ask it to solve that scenario in python.
- B. Paste Scenario and the code that the LLM generated.
- C. Code your own solution for that scenario and identify the differences.

Question 2

- A. Write one single GPT prompt that asks the LLM to generate a code with at least 3 nested conditions, 2 loops, 1 continue and 1 break that has at least 25 lines. Then ask it to determine the CFG graph of that code.
- B. Paste the LLM generated code and CFG.
- C. Mark the lines of code with proper node numbers.
- D. Draw the CFG yourself and identify the differences between the two versions. (You can use draw.io for this or handdraw the graph and insert in the doc)
- E. Calculate the cyclomatic complexity from your CFG.

Answer Template:  Assignment-2 Answer Template

Submission Guideline:

- Create a copy of the template and write your answers accordingly.
- Upload your assignment in PDF format to your google drive.
- Rename the file using this format: ID_Section_Assignment2.pdf
- Go to the share option and update the link to "Anyone Can View"
- Paste the Link in the form
- Submission Form: <https://forms.gle/eDbccySCb9v4VFi59>
- Last date of submission: 3rd January, 2026, 11:59 PM